Binary Classification (2 class classification)

Labels

Predictions

Trainings Approach

Data Split

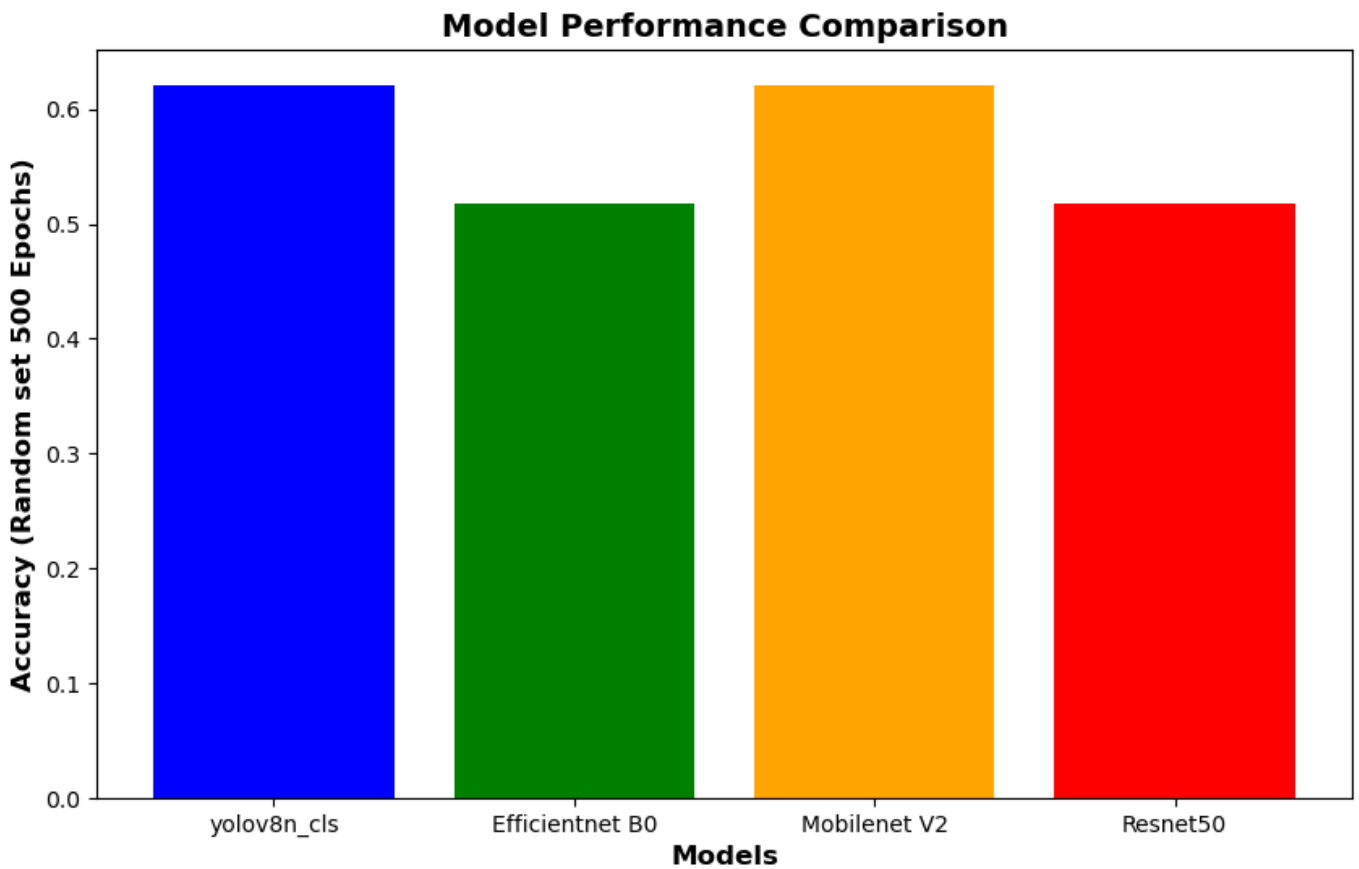
- Training set 760 images
- Validation set 79 images
- Test set 46 pictures
- Pictures from various sources online 46 pictures

3 different training sequences were used

- First: Decision if model is worthwhile, train 150 epochs and reach about 50% accuracy on a random data set.
- Second: Train 500 epochs, if the model improves compared to the 150 epochs the model is chosen.
- Third: Find out the ideal number of epochs.

Results of the second procedure on a random data set

- Yolov8n_cls
- Efficientnet B0
- Mobilenet V2
- Resnet50

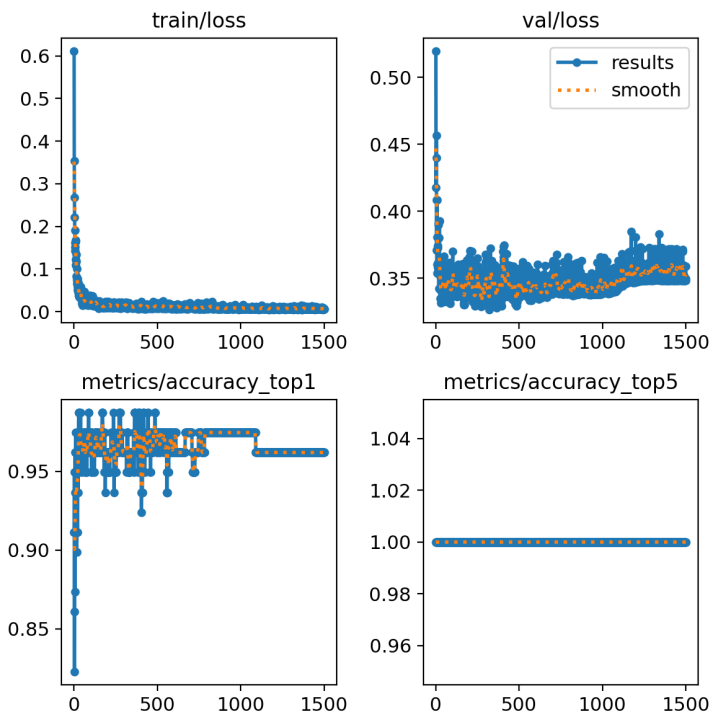


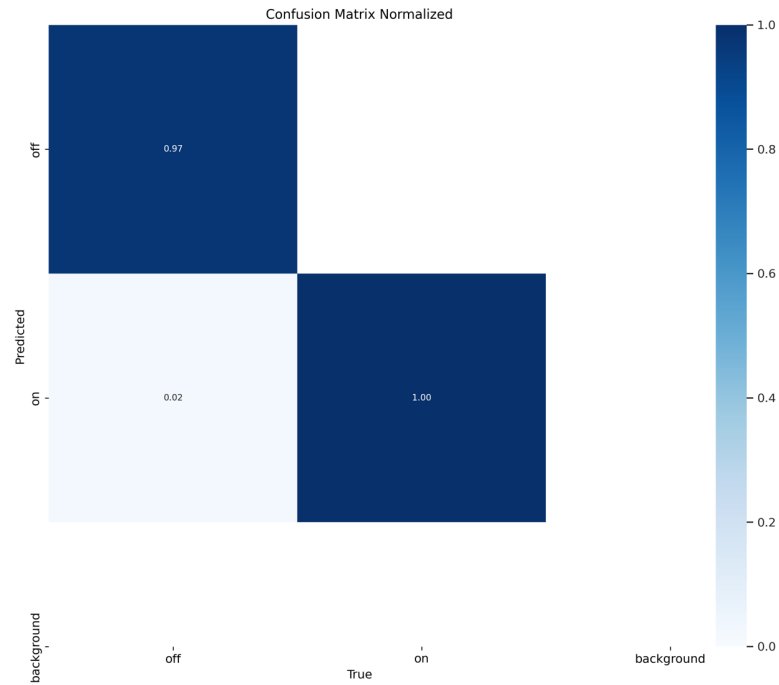
Yolov8n_cls

- Ideal number of epochs to minimize the loss of the validation set is 250 to 750 epochs

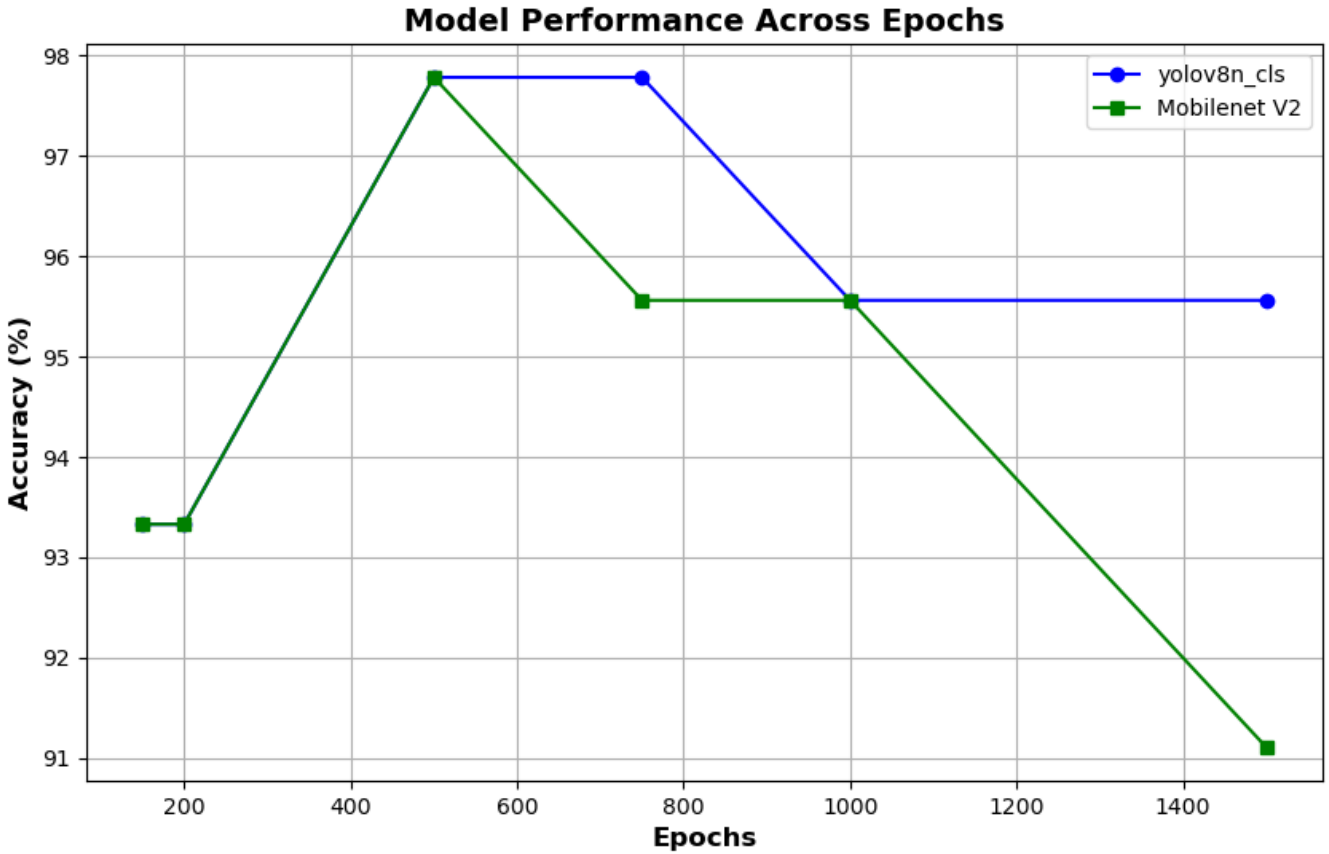
- The training loss is not strongly influenced by overtraining

- Accuracy varies by 2-3% when overtraining





Performance of the two best models on the Test set



Next data to be collected independently of SBB, in public places in ZG & LU

I45, Zug

Pilatusmarkt, Luzern

Expanding Dataset Size through Data Augmentation

Goal: Increase the dataset's size and variety to improve model generalization and robustness.

1. Rotation Augmentation:
   - Introduced slight rotational variations to each image.
   - Simulates real-world variations in orientation.
   - Helps the model generalize better to images with slight rotations.
   - Rotation only by +10° and – 10°; otherwise, it could change the class of the image.
2. Brightness Adjustment:
   - Slightly increases or decreases the brightness in images.
   - Allows the model to adapt to varying lighting conditions, such as day, night, or overcast situations.
3. Contrast Adjustment:
   - Modifies contrast to make certain features either more prominent or more subtle.
   - Improves the model's ability to recognize features under varying contrast levels, enhancing detection under different conditions.

Result:

•**Increased Dataset Size**: Training and validation sets now include both original and augmented images.

Train set: from 760 images to 2280

Validation set: 79 images to 237

•**Better Model Robustness**: The model learns to recognize images in varying conditions, enhancing its generalization to real-world scenarios.

Key Metrics for Model Evaluation on the Mobilenet V2

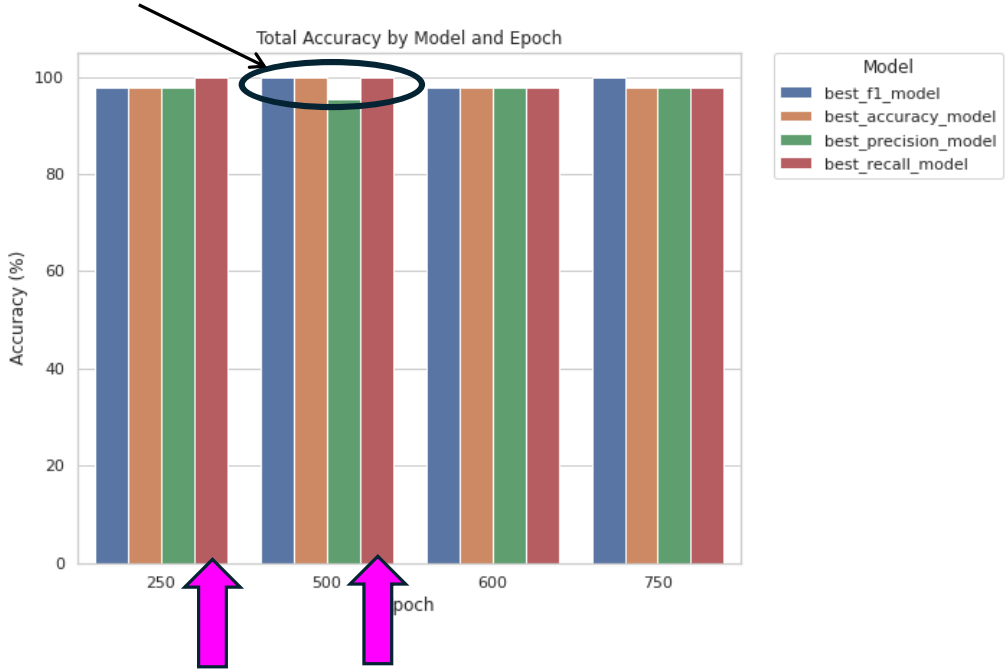
	Precision	Recall	F1 Score
Formula	$\frac{True\ Positives\ (TP)}{TP + False\ Positives\ (FP)}$	$\frac{True\ Positives\ (TP)}{TP + False\ Negatives\ (FN)}$	$2 * \frac{Precision\ * Recall}{TP + False\ \ Negatives(FN)}$
Definition	Measures the percentage of positive predictions that are correct.	Calculates the percentage of actual positives correctly identified by the model.	Harmonic mean of Precision and Recall, balancing the two metrics.
Use Case	**A high precision** reduces "false positives" (where the model mistakenly flags something as "on track" when it isn't), which is useful to avoid unnecessary or costly interventions.	**A high recall** means fewer "false negatives" (i.e., fewer cases where something is on the tracks but goes undetected)	Useful for imbalanced datasets or when both false positives and false negatives matter.

	Accuracy
Formula	$\frac{TP + True\ Negatives(TN)}{Total\ Predictions\ (TP + TN + FP + FN)}$
Definition	Measures the overall percentage of correct predictions.
Use Case	Accuracy gives a general sense of how often the model is correct, it may be less critical here. Since detecting items "on track" is likely a smaller subset of all possible cases, high accuracy alone doesn't guarantee low false negative or false positive rates.

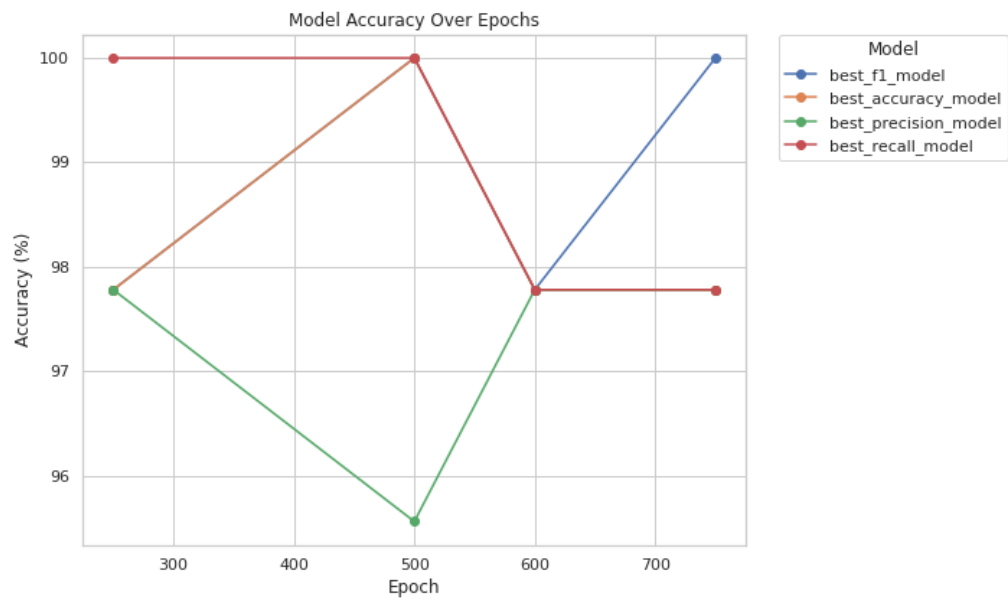
- **Recall** would be the most important aspect of a "fail-safe" approach, as it minimizes the risk of missing the robot completely on the tracks. There's a greater chance that anything actually on the tracks will be detected, minimizing missed detections (false negatives).
- **Precision** would be more important if minimizing false alarms is crucial.
- **F1 Score** is valuable if there's a need to balance both precision and recall for better overall performance.

Assessing the 4 metrics with different Epochs on the Test set

Best on average

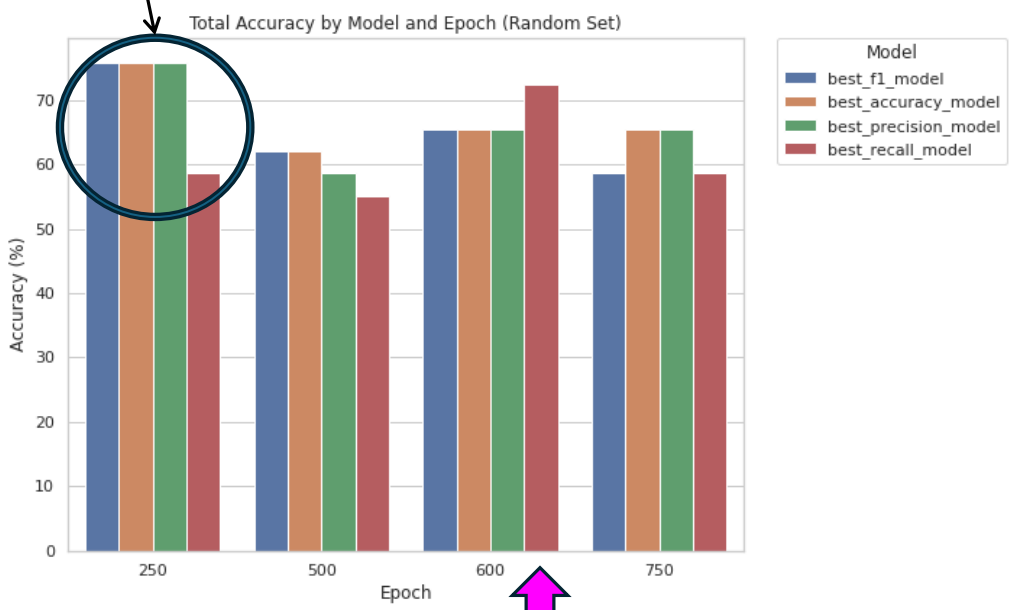


Best for Recall

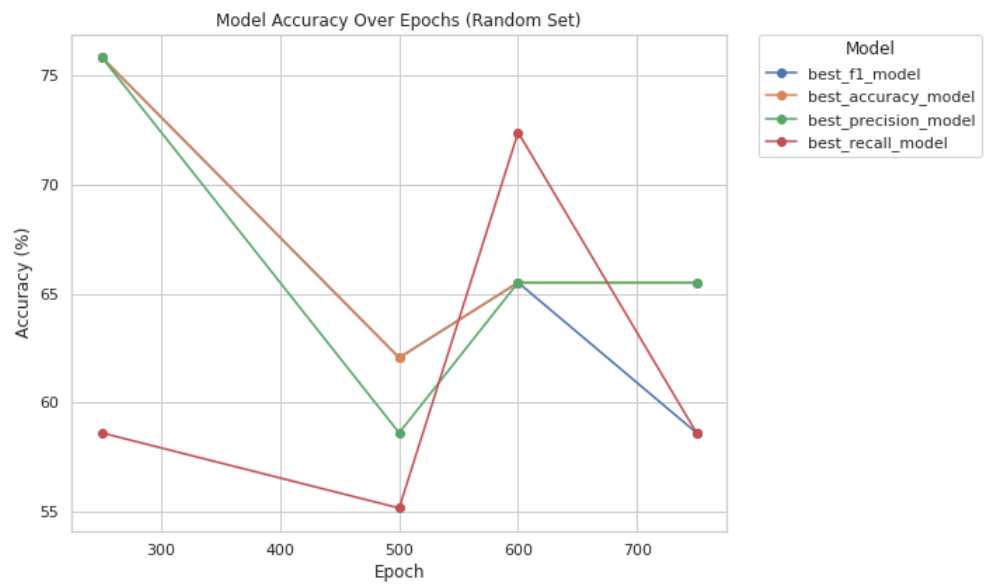


Assessing the 4 metrics with different Epochs on the Random set

Best on average

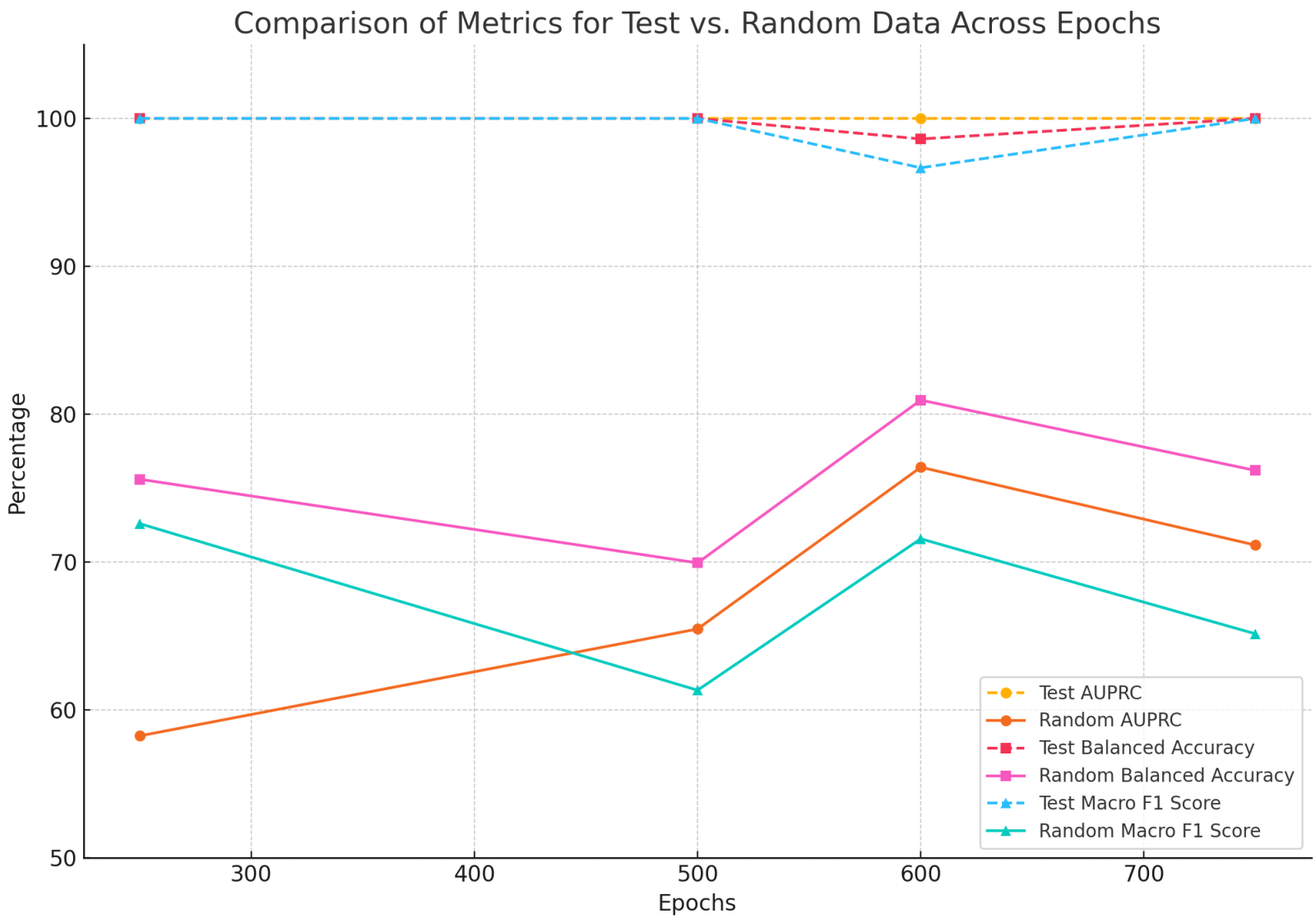


Best for Recall



Taking the Data split in the Test and Random split into consideration

Metric	Balanced Accuracy	Macro F1 Score	AUPRC	Micro F1 Score
Formula	$\frac{(\text{Sensitivity} + \text{Specificity})}{2}$	$\frac{(\text{F1 Score of Class 1} + \text{F1 Score of Class 2})}{2}$	Area under the Precision-Recall curve	$\frac{2*(\text{Micro Precision} * \text{Micro Recall})}{(\text{Micro Precision} + \text{Micro Recall})}$
Definition	Average of recall scores for each class, giving equal weight to each class regardless of its size.	Calculates the average F1 score for each class without considering class size, providing an overall F1 measure.	Represents the area under the precision-recall curve, which reflects the model's ability to distinguish classes.	F1 score calculated by aggregating the contributions of all classes to compute a global measure of precision/recall.
Use Case	Useful for imbalanced datasets where accuracy alone may be misleading.	Ideal for multi-class or imbalanced datasets where each class's performance matters equally.	Useful in imbalanced datasets to better understand the trade-off between precision and recall.	Helpful for multi-class problems with balanced importance across all classes.


Comparison of Metrics for Test vs. Random Data Across Epochs
Percentage
Epochs
100
90
80
70
60
50
300
400
500
600
700
Test AUPRC
Random AUPRC
Test Balanced Accuracy
Random Balanced Accuracy
Test Macro F1 Score
Random Macro F1 Score